

Intro to SQLAlchemy

 <https://github.com/learn-co-curriculum/python-p3-intro-to-sqlalchemy>  <https://github.com/learn-co-curriculum/python-p3-intro-to-sqlalchemy/issues/new>

Learning Goals

- Use an external library to simplify tasks from earlier ORM lessons.
- Manage database tables and schemas without ever writing SQL through Alembic.
- Use SQLAlchemy to create, read, update and delete records in a SQL database.

Key Vocab

- **Schema**: the blueprint of a database. Describes how data relates to other data in tables, columns, and relationships between them.
- **Persist**: save a schema in a database.
- **Engine**: a Python object that translates SQL to Python and vice-versa.
- **Session**: a Python object that uses an engine to allow us to programmatically interact with a database.
- **Transaction**: a strategy for executing database statements such that the group succeeds or fails as a unit.
- **Migration**: the process of moving data from one or more databases to one or more target databases.

Introduction

In our previous module, we learned about **ORMs**. Using `sqlite3` and concepts in object-oriented programming in Python, we built tools to map Python classes to database tables and vice-versa. This is a useful skill for your future work, but it has some distinct limitations:

- It only works with one type of SQL database, `sqlite3`.
- It only covers the functionality that we needed for our `Dog` and `Song` classes.

- We won't know how other groups' ORMs work.
- Others won't know how our ORM works.

In order to address these concerns, several groups of programmers began development on ORMs that had a broad range of compatibilites and functionality, robust documentation, and regular maintenance to keep up with evolving programming languages.

In our curriculum, we will be working with [SQLAlchemy](https://www.sqlalchemy.org/), a popular Python ORM. There are several other popular Python ORMs- most notably the [Django](https://www.djangoproject.com/) ORM- but SQLAlchemy provides us the most functionality without too much that we don't need. It is used by many big tech companies like *Yelp!*, *reddit*, and *DropBox*.

That being said, SQLAlchemy is a very powerful library, and we will only cover a small fraction of its capabilities in this curriculum. There are many fantastic resources available to you for free online or through **Moringa** that will allow you to continue to learn all about SQLAlchemy as you proceed through your career.

Resources

- [Python 3.8.13 Documentation - Python](https://docs.python.org/3.8/)
- [sqlite3 — DB-API 2.0 interface for SQLite databases - Python](https://docs.python.org/3/library/sqlite3.html)
- [SQLAlchemy ORM Documentation - SQLAlchemy](https://docs.sqlalchemy.org/en/14/orm/)
- [SQLAlchemy — Python Tutorial - Towards Data Science](https://towardsdatascience.com/sqlalchemy-python-tutorial-79a577141a91)
- [Alembic 1.8.1 Documentation - SQLAlchemy](https://alembic.sqlalchemy.org/en/latest/)
- [Tutorial \(Alembic\) - SQLAlchemy](https://alembic.sqlalchemy.org/en/latest/tutorial.html)